

5/18

DATE

	RAW SLUDGE	V. RAW SLUDGE	RAW SLUDGE	DIG
	T.S.	T.V.S.	PUMPED	TEMP
1	2.93	77.19	14400.00	97.00
2	3.11	78.33	14400.00	97.00
3	3.19	74.39	14400.00	97.00
4	3.44	78.79	14400.00	96.00
5			14400.00	97.00
6			14400.00	97.00
7	3.03	75.81	14400.00	96.00
8	3.31	79.31	14400.00	95.00
9	3.12	79.69	14400.00	96.00
10	3.18	72.86	14400.00	95.00
11	2.99	75.44	14400.00	95.00
12			14400.00	95.00
13			14400.00	95.00
14	3.02	77.19	14400.00	95.00
15	3.20	75.81	14400.00	95.00
16	3.24	75.81	14400.00	95.00
17	3.09	75.41	14400.00	95.00
18	3.01	76.19	14400.00	95.00
19			14400.00	97.00
20			14400.00	96.00
21	2.70	75.38	14400.00	97.00
22	3.22	75.81	14400.00	97.00
23	2.70	75.86	17600.00	97.00
24	3.00	74.19	14400.00	98.00
25	3.26	72.13	14400.00	98.00
26			14400.00	94.00
27			14400.00	94.00
28			14400.00	98.00
29	2.97	76.47	14400.00	98.00
30	3.10	71.67	14400.00	97.00
31	3.07	73.68	14400.00	95.00
	67.88	1667.41	449600.00	2979.00
	3.09	75.79	14503.23	96.10

May-18		Raw TVS	Digester Temp.
CS Info	AVERAGE>>	75.79	96.10

The above is

May-18	Post TVS (OUT)	
CS Info	AVERAGE>>	58.89
		IN-OUT
		IN-(IN x OUT)

Hauling dates:	Post TVS (OUT)	
8-May	57.69	0.7579 - 0.5889
9-May	61.97	0.7579 - (0.7579 x 0.5889)
10-May	59.60	
11-May	58.62	
12-May	56.58	0.169
13-May	58.90	0.7579 - 0.4463
		0.3116

0.169
0.3116

38% Vc
0.5424

TRANS	SUPER	DECANT
1 - 2	#2	# 3 or # 4
56730.00	10065.00	2400.00
	3695.00	4200.00
	11230.00	
55205.00	1220.00	17400.00
		9600.00
	7625.00	
	5490.00	
51850.00	7930.00	
	9455.00	
	11285.00	
39955.00	15250.00	
	8845.00	
	8540.00	
63440.00	6100.00	
	7930.00	
	6710.00	
	8845.00	
42395.00	5490.00	
	2440.00	
	4880.00	
41175.00	5490.00	
	5490.00	
	3695.00	
34465.00	9760.00	
30500.00	9455.00	
	6710.00	
	3660.00	
28060.00	6100.00	
	9760.00	
19825.00	8845.00	
	7930.00	
463600.00	219920.00	33600.00
42145.45	7330.67	8400.00

10/18
DATE

RAW SLUDGE T.S.
2.28
2.17
2.20
2.14
2.25
2.19
2.16
2.14
2.10
1.80
2.42
2.21
2.33
2.43
2.28
2.28
2.25
2.29
2.35
2.36
2.30
2.41
2.38
51.72
2.25

1
2
3
4
5
6
7
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11
12
13
14
15
16
17
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31

Cindy's information to determine if the WWTP met the 38% reduction of volatile solids at the time of lan

Oct. 2018

CS Info

AVERAGE>>

the Raw TVS, now I look at the document labeled 'biosolids lab data', for is their POST TVS data, collected during

Below is the data POST TVS, collected during the land application events.

Oct. 2018

CS Info

AVERAGE>>

$$\times 100\% =$$
$$\times 100\% =$$
$$\times 100\% =$$
$$= 0.5424$$

Volatile Solids Reduction Met? Y or N = Y

x	100% =	54.24
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Hauling dates:

Post TVS (OUT)

16-Oct

55.80

17-Oct

54.76

18-Oct

58.14

19-Oct

55.00

20-Oct

57.89

V. RAW SLUDGE T.V.S.	RAW SLUDGE PUMPED	DIG TEMP	TRANS 1 - 2	SUPER #2
71.11	14400.00	98.00	19825.00	8845.00
73.33	14400.00	98.00		15555.00
74.74	14400.00	98.00	50630.00	7015.00
73.17	14400.00	98.00		8235.00
76.09	14400.00	98.00	29280.00	14640.00
	14400.00	98.00		10980.00
	14400.00	98.00		10980.00
71.74	14400.00	98.00	47275.00	3965.00
72.09	14400.00	98.00		9760.00
64.10	14400.00	98.00		6405.00
77.27	14400.00	98.00		14335.00
75.00	14400.00	98.00	39650.00	4880.00
	14400.00	99.00		12200.00
	14400.00	98.00		9760.00
68.75	14400.00	98.00	23180.00	3355.00
71.11	14400.00	98.00		10065.00
74.07	14400.00	99.00	17995.00	5795.00
70.59	14400.00	99.00	51545.00	7930.00
70.83	14400.00	98.00		6710.00
	14400.00	98.00		12810.00
	14400.00	98.00		12810.00
72.92	14400.00	98.00	40260.00	5490.00
73.33	14400.00	98.00		6100.00
71.15	14400.00	98.00		11590.00
73.47	14400.00	98.00		5490.00
71.43	14400.00	98.00	53375.00	2745.00
	14400.00	98.00		8540.00
	14400.00	98.00		9760.00
71.74	14400.00	98.00		9150.00
69.39	14400.00	98.00	28060.00	4575.00
72.34	14400.00	98.00		4270.00
1659.76	446400.00	3041.00	401075.00	264740.00
72.16	14400.00	98.10	36461.36	8540.00

d application.	
Raw TVS	Digester Temp.
72.16	98.10

their hauling events.

Post TVS (OUT)

56.32

IN-OUT

IN-(IN x OUT)

X 100% =

0.7216 - 0.5632

0.7216 - (0.7216 x 0.5632)

x 100% =

0.1584

0.7216 - 0.4064

x 100% =

0.3152

0.1584

0.3152

=

0.5025

38% Volatile Solids Reduction Met? Y or N = Y

0.5025

x

100% =

50.25

DECANT
3 or # 4

3600.00

2400.00

4200.00

5400.00

15600.00

3900.00



3/19

4/19

DATE	RAW SLUDGE	V. RAW SLUDGE
	T.S.	T.V.S.
1	3.26	68.18
2		
3		
4	3.27	80.28
5	3.26	77.27
6	3.22	78.26
7	3.16	77.53
8	3.07	78.69
9		
10		
11	3.31	76.12
12	3.10	77.42
13	2.91	76.67
14	3.09	77.42
15	3.57	73.97
16		
17		
18	3.13	74.58
19	2.74	78.69
20	3.27	78.46
21	3.20	78.74
22	3.21	78.26
23		
24		
25	3.02	78.33
26	3.04	76.19
27	3.15	75.81
28	3.03	76.21
29	2.99	79.03
30		
31		
	65.99	1616.10
AVG	3.14	76.96

DATE	RAW SLUDGE	V. RAW SLUDGE
	T.S.	T.V.S.
1	2.89	68.85
2	3.19	77.61
3	3.46	73.53
4	3.13	75.41
5	2.38	76.09
6		
7		
8	3.40	78.79
9	3.18	77.97
10	3.22	77.05
11	3.03	87.93
12	3.35	77.33
13		
14		
15	2.94	80.00
16	3.17	79.03
17	3.17	81.67
18	2.96	78.33
19		
20		
21		
22	2.98	80.88
23	2.84	78.08
24	3.10	77.97
25	2.89	80.00
26	3.03	77.97
27		
28		
29	3.41	70.77
30	3.75	72.73
	65.47	1627.99
AVG	3.12	77.52

Mar-19			Apr-19		
		Raw TVS (IN)			Raw TVS (IN)
CS Info	AVERAGES>>>	76.96	AVERAGES>>>		77.52

The above is t

Mar-19			Apr-19		
		Post TVS (OUT)			Post TVS (OUT)
CS Info	AVERAGES>>>		AVERAGES>>>		

No hauling during this month.

No hauling during this month.

DATE	RAW SLUDGE	V. RAW SLUDGE	RAW SLDGE	DIG	TRANS
	T.S.	T.V.S.	PUMPED	TEMP	1 - 2
1			14400.00	98.00	
2	2.19	76.74	14400.00	98.00	
3	2.27	74.47	14400.00	98.00	39040.00
4	2.26	74.42	14400.00	98.00	
5	2.24	74.42	14400.00	98.00	
6	2.36	72.73	14400.00	98.00	39955.00
7			14400.00	98.00	
8			14400.00	98.00	
9	2.28	73.91	14400.00	98.00	
10	2.55	75.00	14400.00	98.00	
11	2.48	76.60	19200.00	98.00	39955.00
12	2.75	69.64	19200.00	96.00	
13	2.37	74.00	14400.00	96.00	53070.00
14			14400.00	97.00	
15			14400.00	97.00	
16	2.65	77.65	14400.00	98.00	
17	2.75	74.51	19200.00	98.00	36905.00
18	2.75	75.44	14400.00	97.00	49410.00
19	2.82	75.00	14400.00	97.00	
20	2.91	77.05	14400.00	98.00	
21			14400.00	98.00	33550.00
22			14400.00	98.00	
23	2.87	78.33	14400.00	97.00	
24			14400.00	97.00	41175.00
25			19200.00	97.00	
26	2.84	75.93	14400.00	97.00	
27	2.76	74.51	14400.00	98.00	58560.00
28			14400.00	97.00	
29			14400.00	97.00	
30	2.91	77.59	14400.00	97.00	
			451200.00	2925.00	391620.00
	49.01	1427.94	15040.00	97.50	43513.33
	2.58	75.15			

Cindy's information to determine if the WWTP met the 38% reduction

Jun-19	Raw TVS (IN)	Digestor Temps.
AVERAGES>>>	75.15	97.50

he **raw** data for TVS, now I look at the document labeled 'biosolids lab data' for their POST TVS data, co

Below is the data **POST** TVS, collected during the

2019 - June	Post TVS (OUT)
AVERAGES>>>	59.16

IN-OUT
IN-(IN x OUT)

Hauling dates:	Post TVS (OUT)
26-Jun	57.61
27-Jun	58.33
28-Jun	58.75
29-Jun	58.67
30-Jun	63.27
1-Jul	58.33

0.7515 - 0.5916
0.7515 - (0.7515 x 0.5916)
0.7515 - 0.4446
0.1599
0.3069
= 0.521

38% Volatile Solids Reduction		
0.521	X	100%

11/19

SUPER #2	DECANT # 3 or # 4	DATE	RAW SLUDGE T.S.	V. RAW SLUDGE T.V.S.
3965.00		1	2.11	72.09
5185.00		2		
9760.00		3		
10675.00		4	1.93	78.05
10675.00		5	1.93	75.00
10675.00		6	1.98	75.00
	9600.00	7	1.88	74.65
13420.00		8	1.98	72.50
10370.00		9		
6710.00		10		
6405.00		11		
	12000.00	12	1.74	75.68
4880.00	6000.00	13	1.97	75.00
6100.00	6000.00	14	1.69	78.24
7320.00		15	1.97	80.00
11590.00		16		
15250.00		17		
5185.00	4800.00	18	2.06	78.05
10370.00		19	1.86	81.08
15555.00		20	1.95	75.00
13115.00		21	1.77	77.46
12200.00		22	1.96	78.95
14030.00		23		
11285.00		24		
7930.00	4800.00	25	2.06	76.74
4880.00		26	1.66	81.82
12200.00		27	1.83	82.21
7015.00		28		
6405.00		29		
8540.00		30		
261690.00	43200.00		34.32	1387.51
9346.07	7200.00		1.91	77.08

of volatile solids at the time of land application.

Nov. 2019	Raw TVS (IN)
AVERAGES>>>	77.08

collected during their hauling events.

land application events.

Nov. 2019	Post TVS (OUT)
AVERAGES>>>	58.55

X 100% =

= $\frac{0.1599}{0.3069}$

X 100%

Hauling dates:	Post TVS (OUT)
19-Nov	60.63
20-Nov	60.47
21-Nov	56.25
22-Nov	55.56
23-Nov	59.84

Met? Y or N = Y

= 52.1

RAW SLUDGE PUMPED	DIG TEMP	TRANS 1 - 2	SUPER #2	DECANT # 3 or # 4
14400.00	98.00	52155.00	2440.00	7200.00
14400.00	98.00		2135.00	9600.00
14400.00	98.00		6710.00	
16800.00	98.00	41480.00	11590.00	6000.00
16800.00	97.00		5490.00	6000.00
16800.00	97.00			9600.00
19200.00	97.00	10370.00	6710.00	9600.00
19200.00	97.00	58865.00		9600.00
19200.00	98.00		7015.00	10800.00
19200.00	97.00	46055.00	7015.00	6000.00
19200.00	97.00		7930.00	10800.00
14400.00	97.00	55815.00		16200.00
14400.00	97.00		15860.00	4200.00
14400.00	97.00		11590.00	4688.00
14400.00	97.00	57645.00	14640.00	3750.00
14400.00	97.00		5490.00	1875.00
14400.00	97.00		10675.00	
19200.00	97.00	36905.00	19215.00	
19200.00	97.00		13725.00	
19200.00	97.00	33550.00	18910.00	
19200.00	96.00	18605.00	18300.00	
19200.00	95.00	71675.00	1830.00	
19200.00	96.00		4270.00	
19200.00	95.00		9150.00	
19200.00	92.00	45750.00	2440.00	
19200.00	92.00	11285.00	5185.00	
14400.00	92.00	59170.00	11590.00	
19200.00	92.00		8540.00	
19200.00	92.00			
14400.00	92.00	45140.00		
516000.00	2879.00	644465.00	228445.00	115913.00
17200.00	95.97	42964.33	9137.80	7727.53

--

Digester
Temps.
95.97



$$\frac{\text{IN-OUT}}{\text{IN}-(\text{IN} \times \text{OUT})} \times 100\% =$$

$\frac{0.7708 - 0.5855}{0.7708 - (0.7708 \times 0.5855)}$	=	$\frac{0.1853}{0.3195}$
0.7708 - 0.4513		

$\frac{0.1853}{0.3195}$	=	0.5799	X	100%
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38% Volatile Solids Reduction Met? Y or N = Y				
0.5799	X	100%	=	57.99